Debansu Adhikary

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PROFILE

Physics undergraduate working at the intersection of theory and computation to bridge abstract equations with scientific visualisation. Currently focusing on computational physics, Python, and quantum foundations to build interpretable, physically meaningful models.

EDUCATION

2024–2028	**University of Calcutta**	B.Sc. Physics (Hons) *Relevant Coursework: Classical Mechanics, Electromagnetism, Mathematical Methods for Physics, Quantum Mechanics (Intro.), Thermal Physics, Computational & Numerical Methods*	SGPA: 6.724 (current)
2022–2023	**CBSE**	Class XII	Percentage: 82.33%
2020–2021	**CBSE**	Class X	Percentage: 91.33%

PROJECTS & ACADEMIC EXPERIENCE

Summer Internship Program on Quantum Information & Computation

Ramakrishna Mission Vidyamandira, Belur Math, June–July 2026

- Completed an intensive research-focused internship, graduating with 100% attendance and a final academic assessment score of 95%.
- Authored a comprehensive theoretical monograph titled *"Foundations of Quantum Mechanics and Quantum Computation,"* mapping out physical architectures for algorithmic speedups.
- Derived multi-qubit tensor product transformations for quantum teleportation and superdense coding circuits, and modeled open quantum systems via density matrices and spectral decomposition.
- Evaluated phase kickback and quantum parallelism to build mathematical proofs of exponential algorithmic supremacy via the Deutsch and Deutsch-Jozsa protocols.

Asteroid Search and Astrometric Analysis

International Astronomical Search Collaboration, November 2025

- Analysed real astronomical imaging data to identify candidate near-Earth objects through frame-to-frame motion analysis.
- Performed astrometric calibration using reference stars from the Gaia catalog and measured object positions across multiple observations.
- Identified multiple provisional moving-object candidates, documenting detection parameters, observational uncertainty, and potential false positives.
- Gained experience with the practical limitations and preliminary nature of observational asteroid detection workflows.

Summer School on Quantum Foundations

Bhaktivedanta Institute & C-DAC, June 2025

- Participated in interdisciplinary lectures and discussions on the conceptual and mathematical foundations of quantum mechanics.

- Delivered a theoretical presentation examining interpretational questions in quantum theory, emphasising structured reasoning, conceptual clarity, and connections to standard quantum formalism.

COURSES & CERTIFICATIONS

- **Quantum Information and Emerging Quantum Technologies**
  National Institute of Technology, Rourkela — Jan 2026
- **Introduction to Quantum Natural Language Processing**
  CDAC, Kolkata — Sep 2025
- **Academic Seminars and Colloquia**
  - Prof. M. R. Gupta Memorial National Seminar on Nonlinear and Complex Phenomena
    Jadavpur University, Kolkata — Aug 2025
  - Physics Colloquium: *The Arrow of Time and Black Hole Mystery*
    LNM Institute of Information Technology — Aug 2025

SKILLS

Technical Skills: Python (NumPy, Pandas, Matplotlib), Regression Modeling, Time-Series Analysis, Statistical Data Analysis, Astrometric Analysis, GnuPLOT, LaTeX, Astrometrica

Research Skills: Data Interpretation, Error Analysis, Scientific Visualization, Model Evaluation and Validation

Languages: English, Hindi, Bengali